Comprehensive Roadmap: Q1 Journal Publication Strategy

Parkinson's Disease Detection via Adaptive Tree-Spline Fusion of LightGBM & KAN

Respected Sir, after extensive benchmarking and iterative refinement based on your feedback, we have established a rigorous research methodology. This document presents our finalized 8-phase roadmap designed to produce a publication-ready manuscript for a Q1-tier medical informatics journal (e.g., Computers in Biology and Medicine, Expert Systems with Applications, Artificial Intelligence in Medicine). Each phase addresses a specific reviewer expectation.

Verified Results So Far (Patient-wise GroupKFold, Zero Data Leakage)

Turkish UCI Dataset (756 samples, 252 patients, 754 features):

Rank	Model	Accuracy
1	**Adaptive Fusion (Ours)**	**85.18%**
2	LightGBM	84.26%
3	CatBoost	82.54%
4	XGBoost	82.01%
5	KAN	81.86%
6	SAINT	80.41%
7	NODE	79.88%
8	GATE	79.87%
9	TabTransformer	75.91%
10	TabNet	71.96%

Bengali BenSParX Dataset (900 samples, 120 patients, 71 features):

Rank	Model	Accuracy
1	**Adaptive Fusion (Ours)**	**93.56%**
2	LightGBM	92.56%
3	KAN	91.33%

Phase 1: Feature Selection & Deep Statistical Analysis

- The Turkish dataset contains 754 features, many of which act as noise and degrade deep learning performance. We will perform robust Feature Selection (ANOVA F-test, Mutual Information, and SHAP-based selection) to isolate the top most clinically significant features.
- We will conduct rigorous statistical validations (Welch's t-tests, Mann-Whitney U tests) and generate correlation matrices to demonstrate a profound understanding of the dataset's acoustic properties.
- Goal: Identify the minimal optimal feature subset that maximizes clinical discriminability.

Phase 2: True Clinical Validation via Patient-Isolated Splitting

- Many existing papers suffer from 'Data Leakage' by randomly splitting recordings, artificially inflating accuracy to 90%+ on the UCI dataset.
- We permanently address this critical flaw by utilizing strict Patient-wise Group K-Fold Cross Validation across ALL testing, ensuring no patient's recordings overlap between training and testing sets.

- This directly addresses the most common criticism from Q1 reviewers in this domain and proves true real-world clinical generalizability.

Phase 3: Adaptive Tree-Spline Weighted Fusion (Core Architectural Novelty)

- We propose a novel hybrid architecture fusing LightGBM (gradient-boosted decision trees, excelling at discrete/tabular patterns) with KAN (Kolmogorov-Arnold Networks, excelling at learning continuous nonlinear spline functions).
- Instead of naive 50-50 Soft Voting, we introduce Adaptive Weighted Fusion where the optimal weight ratio dynamically adapts to each dataset's feature dimensionality.
- Verified Result: On the 754-feature Turkish dataset, a 60-40 (LightGBM-KAN) ratio achieves 85.18%, beating standalone LightGBM (84.26%). On the 71-feature Bengali dataset, a natural 50-50 ratio achieves 93.56%, beating standalone LightGBM (92.56%).
- This 'Tree-Spline' complementarity is a strong novelty claim: trees capture discrete splits while splines capture smooth nonlinear vocal cord dynamics.

Phase 4: Explainable AI & Clinical Interpretability (XAI)

- Medical journals strictly require models to be clinically transparent and interpretable.
- We will integrate SHAP (SHapley Additive exPlanations) to visualize exactly which voice features (e.g., Jitter, Shimmer, HNR, MFCC) are driving predictions.
- We will extract the symbolic mathematical splines learned by the KAN model and map them to physical vocal cord impairments observed in Parkinson's patients, effectively removing the 'black box' nature of deep learning.
- This dual-explainability (SHAP for LightGBM + Symbolic Splines for KAN) is a unique contribution not found in existing Parkinson's detection literature.

Phase 5: Global Multi-Dataset Consolidation & Cross-Lingual Validation

- To maximize the paper's scientific impact, we will not limit validation to just two datasets.
- We will perform rigorous testing on multiple available global datasets (e.g., Turkish UCI, Bengali BenSParX, PC-GITA Spanish, Oxford Parkinson's, Italian PD, etc.).
- Two types of validation will be performed:
  (a) Within-Dataset: Patient-wise GroupKFold on each dataset individually.
  (b) Cross-Dataset: Train on Dataset A, Test on Dataset B (and vice versa) to prove language-independent and recording-environment-independent robustness.
- This multi-lingual, multi-corpus consolidation is extremely rare in PD detection literature and will be a powerful differentiator for Q1 acceptance.

Phase 6: Ablation Study (Reviewer-Demanded)

- Q1 reviewers almost always demand an Ablation Study to prove each component's contribution.
- We will systematically measure the impact of removing each component:
  (a) Fusion without Feature Selection vs. with Feature Selection.
  (b) Fusion without Adaptive Weights (naive 50-50) vs. with Adaptive Weights.

(c) LightGBM alone vs. KAN alone vs. Fusion.

(d) Random Split vs. Patient-wise GroupKFold (to quantify the leakage inflation).

- This table will conclusively prove that every component of our framework adds measurable value.

Phase 7: Statistical Significance Testing

- Simply showing higher accuracy is not enough for Q1 journals. We must prove the improvement is statistically significant, not due to random chance.
- We will perform McNemar's test (comparing prediction disagreements between Fusion vs. each baseline) and Wilcoxon signed-rank test (comparing fold-wise accuracy distributions).
- We will report p-values and confidence intervals alongside all accuracy claims.
- This statistical rigor is what separates a Q1-accepted paper from a Q2/Q3 rejection.

Phase 8: Comprehensive Paper Drafting & Presentation

- Produce a polished proposed architecture diagram (block diagram) showing the complete pipeline: Raw Audio -> Feature Extraction -> Feature Selection -> Parallel LightGBM & KAN training -> Adaptive Weighted Fusion -> Final Prediction + Explainability.
- Include comprehensive comparison tables against state-of-the-art methods from recent literature (2022-2025) on the same datasets, ensuring our method is positioned favorably.
- Prepare high-quality figures: ROC curves, Confusion Matrices, SHAP beeswarm plots, KAN symbolic spline visualizations, and training convergence curves.
- Draft the manuscript following the target journal's template with all required sections: Introduction, Related Work, Methodology, Experiments, Results, Discussion, and Conclusion.

Summary of Novelty Claims (For Reviewer Response)

N1: First application of Kolmogorov-Arnold Networks (KAN) for Parkinson's disease detection from speech signals.

N2: Novel Adaptive Tree-Spline Fusion architecture combining gradient boosting with learnable spline functions, with mathematically optimized dataset-specific weight ratios.

N3: Dual Explainability framework - SHAP for tree-based reasoning + symbolic spline extraction from KAN for continuous feature interpretation.

N4: Rigorous zero-leakage evaluation using Patient-wise GroupKFold, directly exposing the inflated results reported in prior works.

N5: First multi-lingual, cross-corpus validation for PD speech detection across Turkish, Bengali, Spanish, and other datasets, proving language-independent generalization.

Conclusion

By systematically executing these eight phases - eliminating data leakage, introducing Adaptive Tree-Spline Fusion, providing dual medical explainability, validating across multiple global datasets, conducting rigorous ablation and statistical significance testing, and presenting state-of-the-art comparisons - our framework will be statistically bulletproof and highly competitive for publication in a Q1 medical informatics journal.